

Titanic Prediction: analysis and report

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This is a report and analysis of the neural network I developed myself in Python and R using PyTorch, NumPy, pandas, and other libraries. The object of the network is to successfully predict who would survive the Titanic based on age, sex and other factors. My network achieved a 78.32% accuracy on unseen testing data, and 86.38% in general across the whole dataset, which is a good score comparable to other machine learning models developed by others.

The data and python scripts are available for viewing and your own testing in the GitHub repository, the R script is also in the repository, as well as below along with my comments and explanations.

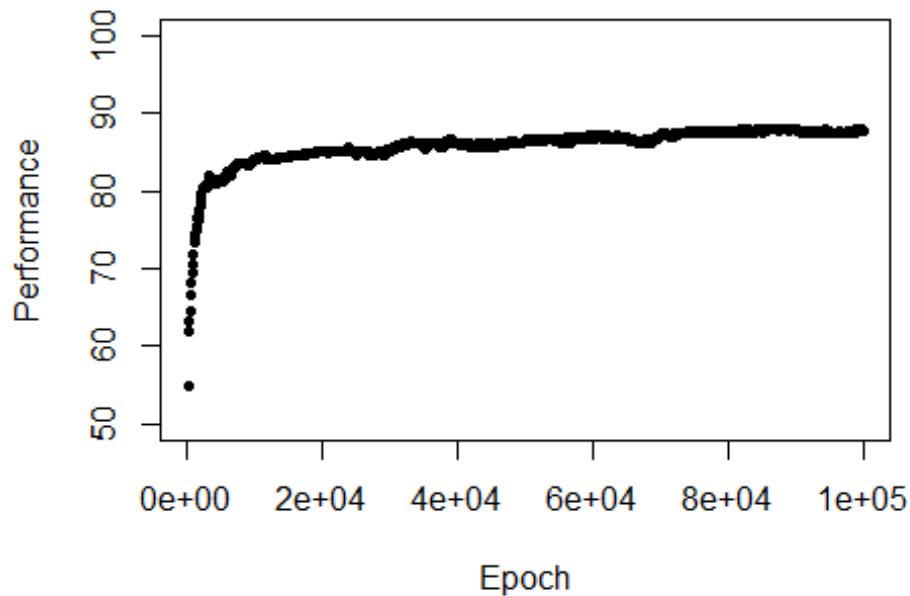
```
library(ggplot2)
library(readxl)
library(patchwork)
pass_info = read.csv("train.csv", header = TRUE)
survived = read.csv("survivaldata.csv", header = TRUE)
model_predictions = read_excel(path = "predictions.xlsx", range="B1:C713")
true_data = cbind(pass_info, survived)

true_data$PassengerId = NULL
true_data$Age = true_data$Age + 30
true_data$Fare = true_data$Fare + 35
true_data$Sex = ifelse(true_data$Sex == 1, "Male", "Female")
true_data$Pclass = true_data$Pclass + 2
true_data$Survived = ifelse(true_data$Survived == 1, "Survived", "Died")
true_data$Embarked = ifelse(
  true_data$Embarked == -1, "Southampton", ifelse(
    true_data$Embarked == 0, "Cherbourg", ifelse(
      true_data$Embarked == 1, "Queenstown", NA
    )
  )
))

model_predictions$Prediction = ifelse(model_predictions$Prediction == 1, "Survived", "Died")
combined_data = cbind(true_data, model_predictions)
combined_data$PassengerId = NULL

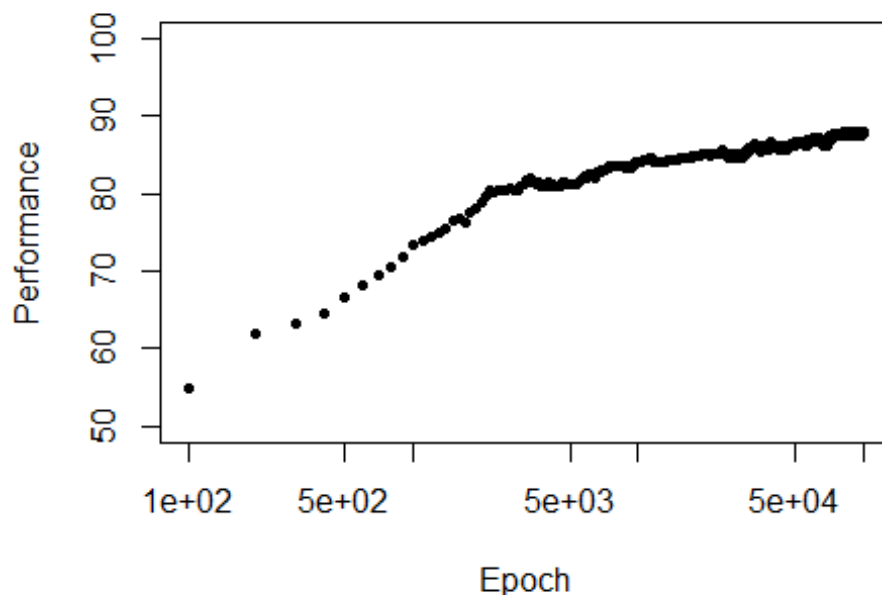
performance = read_excel(path = 'performance.xlsx', range="B1:C1001")
plot(performance*100, pch = 20, ylim=c(50,100))
```

Above I am simply preparing the data for analysis, as well as combining the true data and my model's predicted outcomes into one dataset for ease of comparison.



Above you can see the performance (accuracy) of the model after each epoch (pass through the data), however the interesting part of this graph is the sharp increase at the far left of the graph, hence we can use a log graph to see this more clearly:

```
plot(performance*100, log="x", pch = 20, ylim=c(50,100))
```



Above you can see the log graph, the log scale helps highlight the important rise in performance early on, as well as the slow climb toward higher performance later.

```
confusion_matrix = table(  
  Actual = combined_data$Survived,  
  Predicted = combined_data$Prediction  
)
```

```
##               Predicted  
## Actual      Died Survived  
##   Died       385      39  
##   Survived   58      230
```

I have enlarged this part of the output as I think it is worth looking at, you can see the number of true positives, true negatives, false positives and false negatives, from these we can calculate some interesting performance metrics, see below:

```

TN = confusion_matrix["Died", "Died"]
FP = confusion_matrix["Died", "Survived"]
FN = confusion_matrix["Survived", "Died"]
TP = confusion_matrix["Survived", "Survived"]

accuracy = (TP + TN) / sum(confusion_matrix)
precision = TP / (TP + FP)
recall = TP / (TP + FN)
F1 = 2 * precision * recall / (precision + recall)

```

accuracy

```
## [1] 0.863764
```

precision

```
## [1] 0.8550186
```

recall

```
## [1] 0.7986111
```

F1

```
## [1] 0.8258528
```

These are some common performance metrics used for neural networks.

Accuracy: What proportion of predictions were correct?

Precision: Of everyone the model predicted would survive, what proportion actually survived?

Recall: Of all the people who actually survived, what proportion did the model correctly identify?

F1: Combined score of precision and recall

Tests for survival based on sex

```

p1 = ggplot(combined_data, aes(x = Sex, fill = Survived)) +
  geom_bar(position = "fill") +
  labs(
    title = "Actual",

```

```

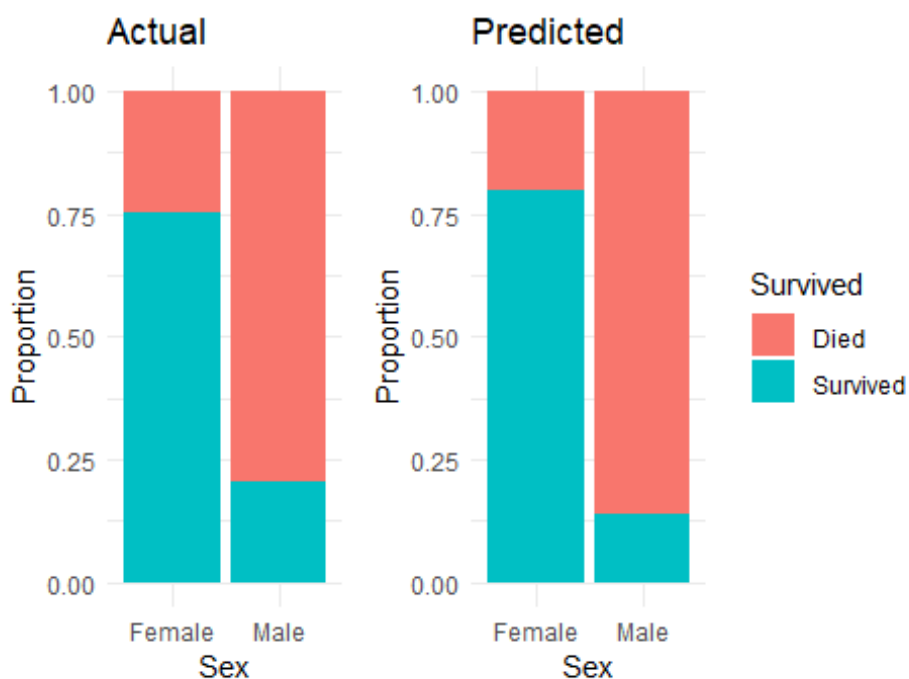
    x = "Sex",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal()

p2 = ggplot(combined_data, aes(x = Sex, fill = Prediction)) +
  geom_bar(position = "fill") +
  labs(
    title = "Predicted",
    x = "Sex",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal()

p1 + p2 +
  plot_layout(ncol = 2, guides = "collect") +
  plot_annotation(title = "Actual vs Predicted Survival by Sex")

```

Actual vs Predicted Survival by Sex



Above you can see survival rates by sex. The plot on the left shows the actual proportions, and the plot on the right shows the proportions my model predicted. As you can see, they are practically identical, with my model slightly predicting too many

female survivors and too few male survivors, but it is extremely close. This is indicative of good model performance. But how much does sex influence survival? See below:

```
chisq.test(table(combined_data$Sex, combined_data$Survived))$p.value * 100
## [1] 4.939417e-44%

chisq.test(table(combined_data$Sex, combined_data$Prediction))$p.value * 100
## [1] 3.060231e-66%
```

Above are the chi-squared tests, the first testing whether sex is independent of survival, we will use a significance level of 1%, we get a p-value of 4.939417e-44%, far below this, hence sex is not independent of survival. The second test is on the model's predictions rather than the actual data, the similarly low p-value shows that the model's predictions are not independent of sex, as we would expect. The rest of this report will follow a similar structure of a graph of actual vs predicted survival rates for some field, and then a chi-squared test.

Tests for survival based on age

```
combined_data$AgeGroup = cut(
  combined_data$Age,
  breaks = c(0, 5, 10, 20, 40, 60, Inf),
  labels = c("0-5", "5-10", "10-20", "20-40", "40-60", "60+"),
  include.lowest = TRUE
)

p14 = ggplot(combined_data, aes(x = AgeGroup, fill = Survived)) +
  geom_bar(position = "fill") +
  labs(
    title = "Actual",
    x = "Age",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
```

```

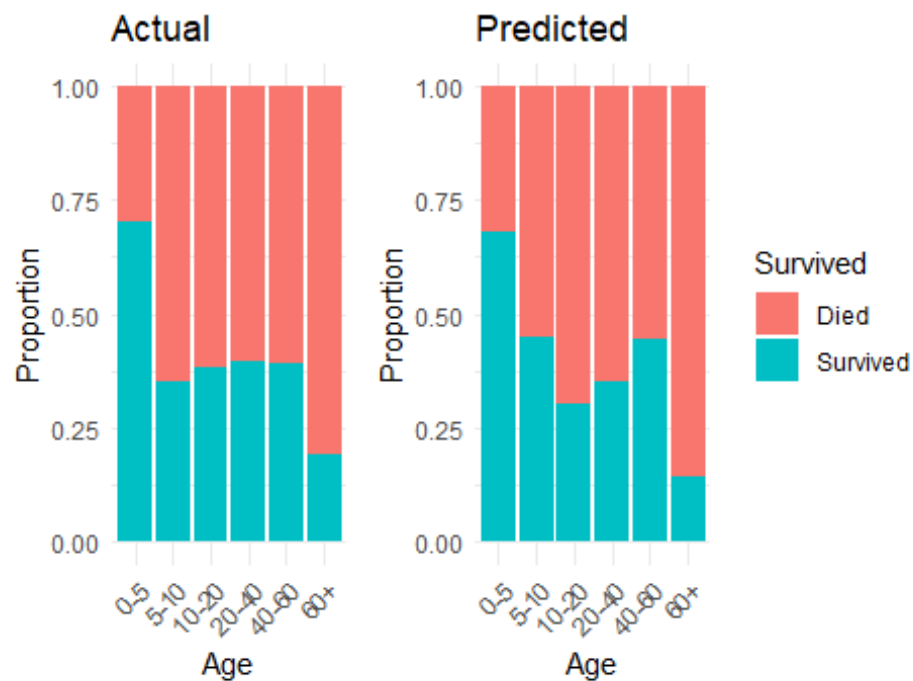
)

p15 = ggplot(combined_data, aes(x = AgeGroup, fill = Prediction)) +
  geom_bar(position = "fill") +
  labs(
    title = "Predicted",
    x = "Age",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

p14 + p15 +
  plot_layout(ncol = 2, guides = "collect") +
  plot_annotation(title = "Actual vs Predicted Survival by Age")

```

Actual vs Predicted Survival by Age



```

chisq.test(table(combined_data$AgeGroup, combined_data$Survived))$p.value * 100
## [1] 0.0763616%

chisq.test(table(combined_data$AgeGroup, combined_data$Prediction))$p.value * 100

```

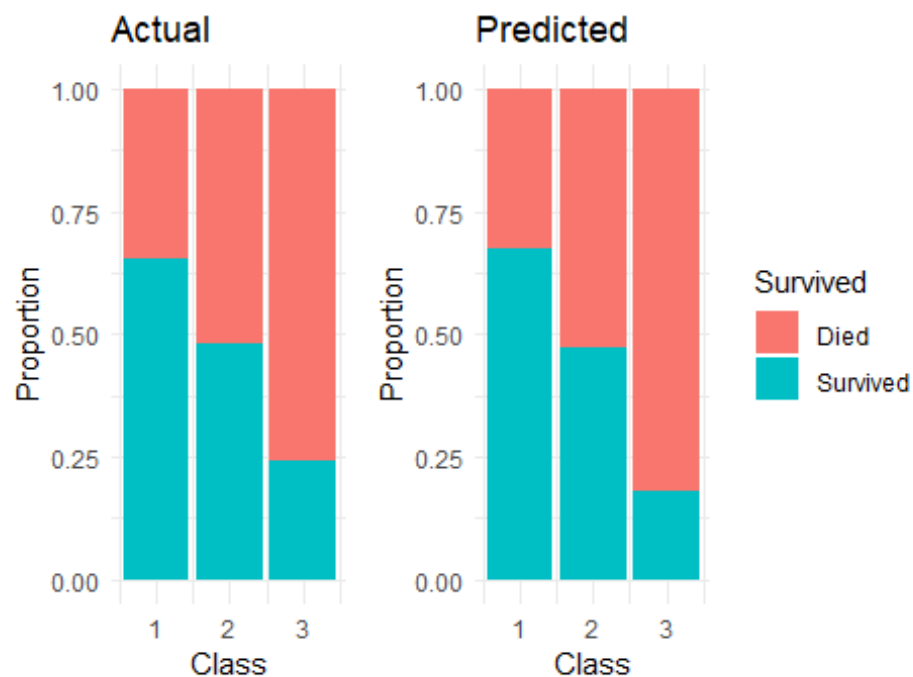
```
## [1] 0.00240307%
```

Above I have grouped the ages, this is so that I can create a bar chart, and so that a chi-squared test can be performed. You can see that our predicted chart doesn't quite follow the actual data quite so closely this time, so this is an area that our model could maybe improve on. Our low p-values above show that survival is dependent on age, in both the actual data and our predicted survival outcomes.

```
### Tests for survival based on passenger class ###
```

```
p3 = ggplot(combined_data, aes(x = Pclass, fill = Survived)) +  
  geom_bar(position = "fill") +  
  labs(  
    title = "Actual",  
    x = "Class",  
    y = "Proportion",  
    fill = "Survived"  
  ) +  
  theme_minimal()  
  
p4 = ggplot(combined_data, aes(x = Pclass, fill = Prediction)) +  
  geom_bar(position = "fill") +  
  labs(  
    title = "Predicted",  
    x = "Class",  
    y = "Proportion",  
    fill = "Survived"  
  ) +  
  theme_minimal()  
  
p3 + p4 +  
  plot_layout(ncol = 2, guides = "collect") +  
  plot_annotation(title = "Actual vs Predicted Survival by Passenger Class")
```

Actual vs Predicted Survival by Passenger Class



```
chisq.test(table(combined_data$Pclass, combined_data$Survived))$p.value * 100
```

```
## [1] 1.667506e-18%
```

```
chisq.test(table(combined_data$Pclass, combined_data$Prediction))$p.value * 100
```

```
## [1] 2.860406e-28%
```

Tests for survival based on siblings or spouses onboard

```
combined_data$SibSpGroup = cut(
  combined_data$SibSp,
  breaks = c(-1, 0, 1, 2, Inf),
  labels = c("0", "1", "2", "3+" ),
  include.lowest = TRUE
)
```

```
p5 = ggplot(combined_data, aes(x = SibSpGroup, fill = Survived)) +
  geom_bar(position = "fill") +
  labs(
    title = "Actual",
    x = "No. of siblings or spouses",
    y = "Proportion",
    fill = "Survived"
  ) +
```

```

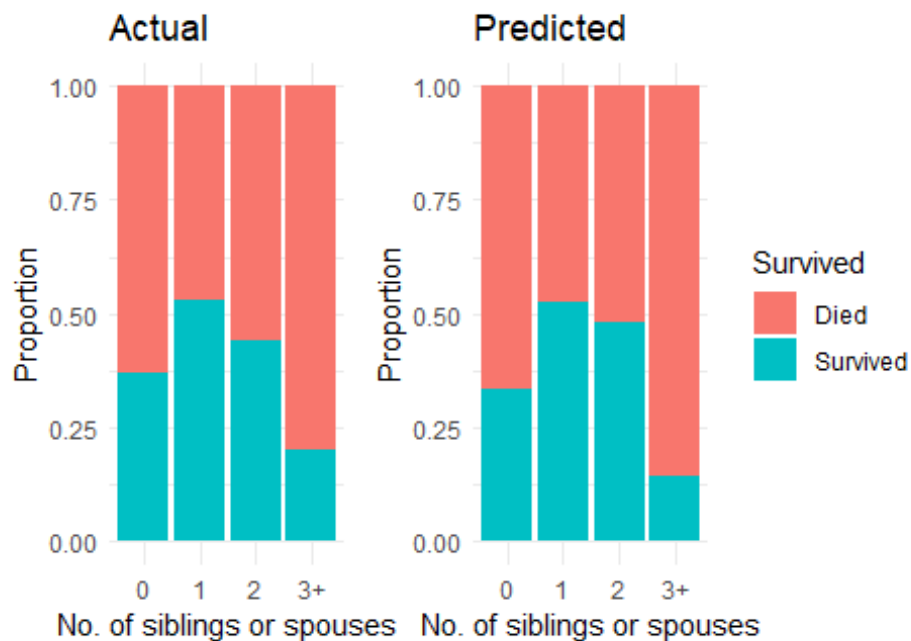
theme_minimal()

p6 = ggplot(combined_data, aes(x = SibSpGroup, fill = Prediction)) +
  geom_bar(position = "fill") +
  labs(
    title = "Predicted",
    x = "No. of siblings or spouses",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal()

p5 + p6 +
  plot_layout(ncol = 2, guides = "collect") +
  plot_annotation(title = "Actual vs Predicted Survival by number of siblings
\nor spouses onboard")

```

Actual vs Predicted Survival by number of siblings or spouses onboard



```

chisq.test(table(combined_data$SibSpGroup, combined_data$Survived))$p.value *
100

## [1] 0.01241349%

chisq.test(table(combined_data$SibSpGroup, combined_data$Prediction))$p.value
* 100

## [1] 0.0001267289%

```

Tests for survival based on parents or children onboard

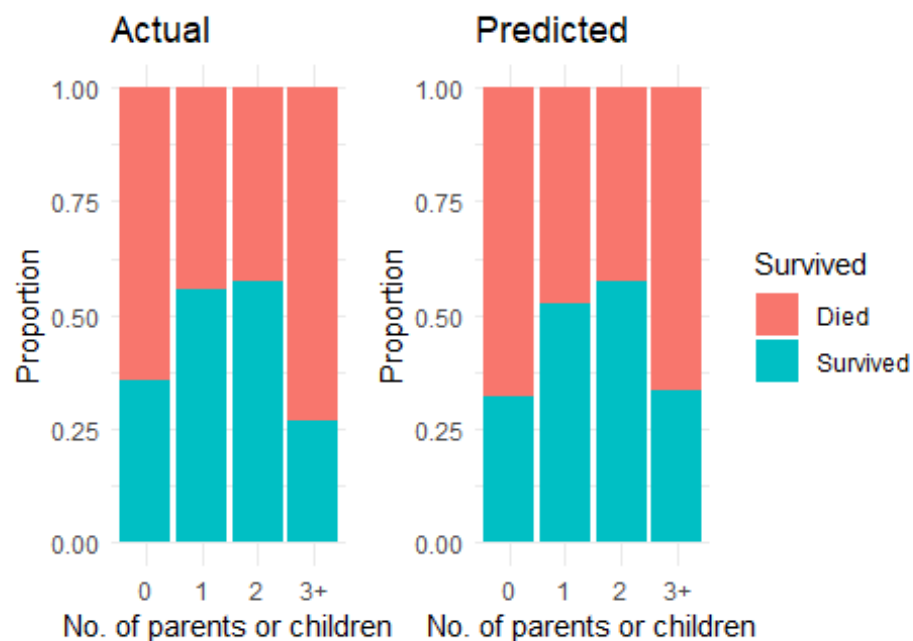
```
combined_data$ParchGroup = cut(  
  combined_data$Parch,  
  breaks = c(-1, 0, 1, 2, Inf),  
  labels = c("0", "1", "2", "3+"),  
  include.lowest = TRUE  
)
```

```
p7 = ggplot(combined_data, aes(x = ParchGroup, fill = Survived)) +  
  geom_bar(position = "fill") +  
  labs(  
    title = "Actual",  
    x = "No. of parents or children",  
    y = "Proportion",  
    fill = "Survived"  
  ) +  
  theme_minimal()
```

```
p8 = ggplot(combined_data, aes(x = ParchGroup, fill = Prediction)) +  
  geom_bar(position = "fill") +  
  labs(  
    title = "Predicted",  
    x = "No. of parents or children",  
    y = "Proportion",  
    fill = "Survived"  
  ) +  
  theme_minimal()
```

```
p7 + p8 +  
  plot_layout(ncol = 2, guides = "collect") +  
  plot_annotation(title = "Actual vs Predicted Survival by number of parents\  
nor children onboard")
```

Actual vs Predicted Survival by number of parents or children onboard



```
chisq.test(table(combined_data$ParchGroup, combined_data$Survived))$p.value *
100
```

```
## [1] 0.001612234%
```

```
chisq.test(table(combined_data$ParchGroup, combined_data$Prediction))$p.value
* 100
```

```
## [1] 0.0002725563%
```

Tests for survival based on embarked port

```
p9 = ggplot(combined_data, aes(x = Embarked, fill = Survived)) +
  geom_bar(position = "fill") +
  labs(
    title = "Actual",
    x = "Embarked",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

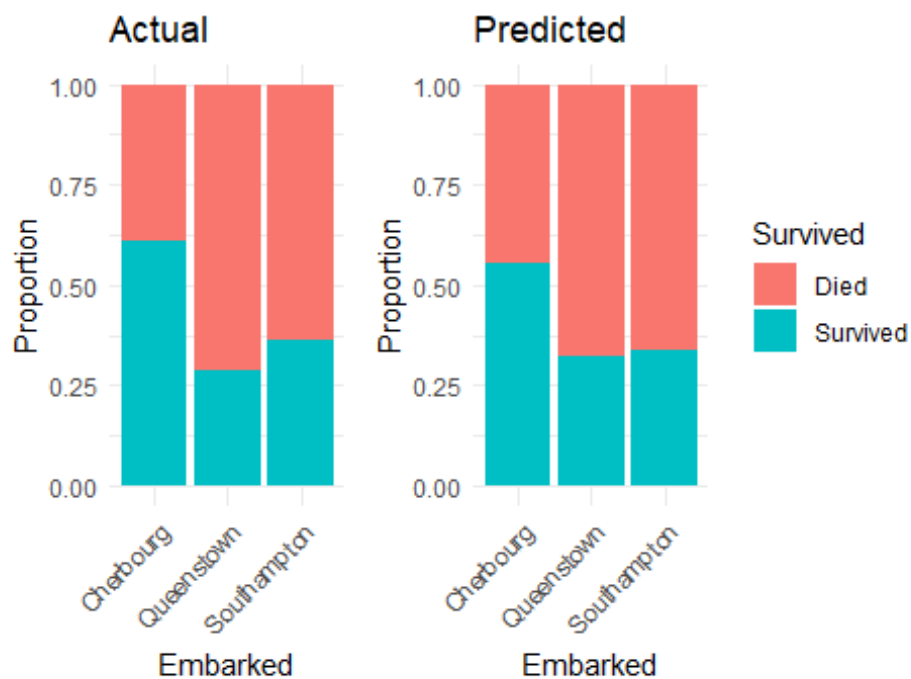
```

p10 = ggplot(combined_data, aes(x = Embarked, fill = Prediction)) +
  geom_bar(position = "fill") +
  labs(
    title = "Predicted",
    x = "Embarked",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

p9 + p10 +
  plot_layout(ncol = 2, guides = "collect") +
  plot_annotation(title = "Actual vs Predicted Survival by Embarked Port")

```

Actual vs Predicted Survival by Embarked Port



```

chisq.test(table(combined_data$Embarked, combined_data$Survived))$p.value * 100
## [1] 8.660307e-05%

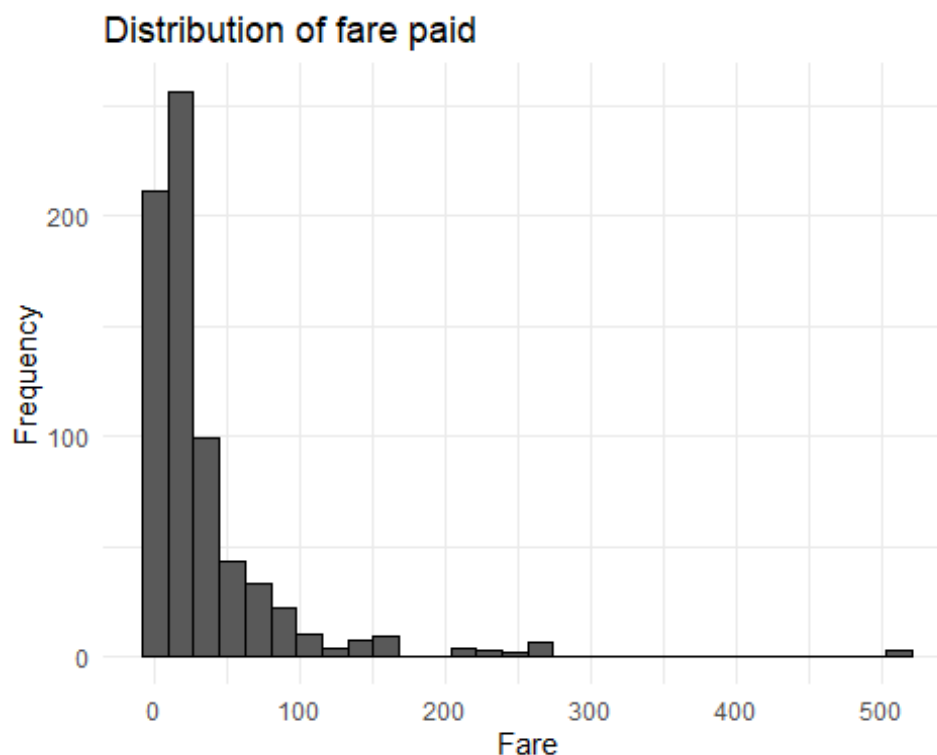
chisq.test(table(combined_data$Embarked, combined_data$Prediction))$p.value * 100
## [1] 0.002750486%

```

```
### Tests for survival based on fare ###
```

```
combined_data$FareGroup = cut(  
  combined_data$Fare,  
  breaks = c(0, 10, 25, 50, 100, 999999),  
  labels = c("£0-10", "£10-25", "£25-50", "£50-100", "£100+"),  
  include.lowest = TRUE  
)  
  
p11 = ggplot(combined_data, aes(x = Fare)) +  
  geom_histogram(colour="black", position = "identity", bins = 30) +  
  labs(  
    title = "Distribution of fare paid",  
    x = "Fare",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```

p11



The above histogram simply shows the distribution of fare paid (£) this is simply to put the next graph in context, rather than to say anything about the performance of the model.

```

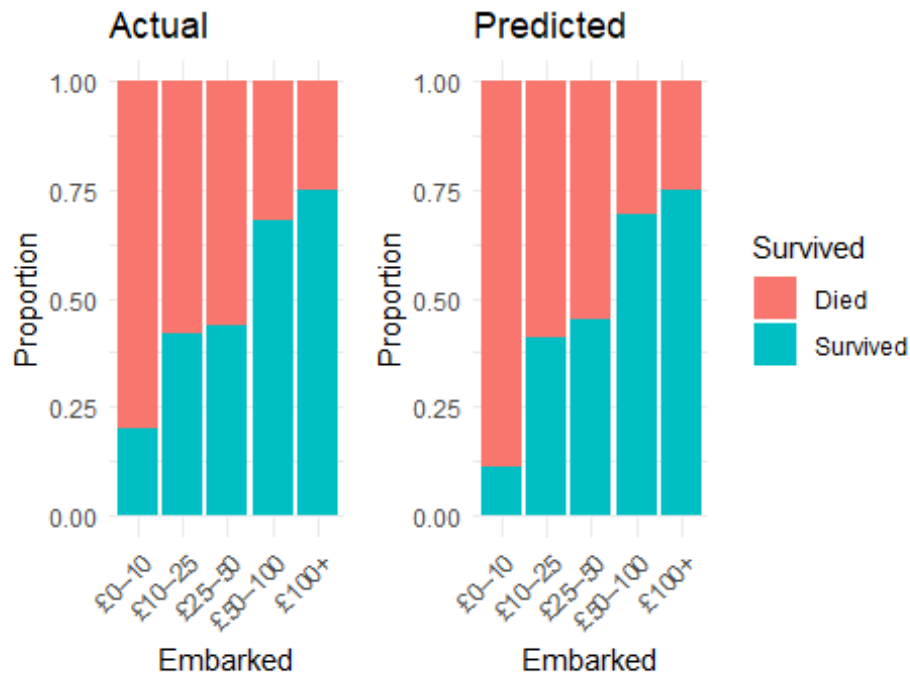
p12 = ggplot(combined_data, aes(x = FareGroup, fill = Survived)) +
  geom_bar(position = "fill") +
  labs(
    title = "Actual",
    x = "Embarked",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal()+
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

p13 = ggplot(combined_data, aes(x = FareGroup, fill = Prediction)) +
  geom_bar(position = "fill") +
  labs(
    title = "Predicted",
    x = "Embarked",
    y = "Proportion",
    fill = "Survived"
  ) +
  theme_minimal()+
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

p12 + p13 +
  plot_layout(ncol = 2, guides = "collect") +
  plot_annotation(title = "Actual vs Predicted Survival by Fare")

```

Actual vs Predicted Survival by Fare



```
chisq.test(table(combined_data$FareGroup, combined_data$Survived))$p.value * 100
```

```
## [1] 1.83358e-17%
```

```
chisq.test(table(combined_data$FareGroup, combined_data$Prediction))$p.value * 100
```

```
## [1] 1.141405e-27%
```

Overall, the model performed well, validating my design choices and provided an interesting insight into the survival statistics of the Titanic. The model could perhaps improve if it was given the names of the passengers, families may have similar survival outcomes, this could lead to a higher accuracy etc. However, overall, the model was given the main statistics that drive survival and performed well.

See my Github for other projects, such as neural networks built from scratch, and a handwriting digit recognition network.